Modeling Ontologies with LLMs

Knowledge Representation and Knowledge Extraction
University of Bologna - 14.04.2025

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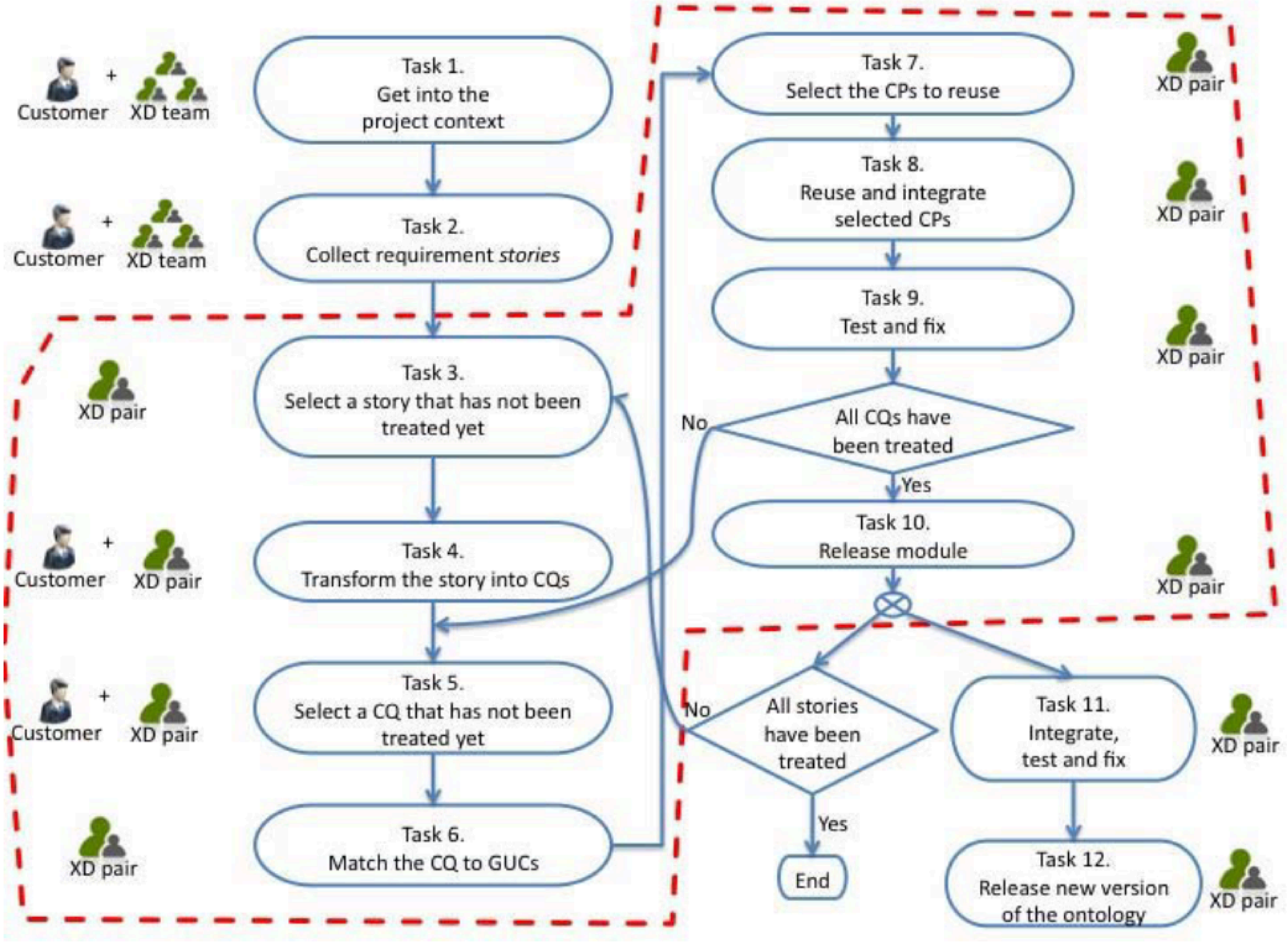

EXTREME DESIGN METHODOLOGY

Context and Literature State of the Art

Large Language Models

Ontology Design Patterns

Words as frame semantic triggers

New modules integration and alignment

MODELING FRAMEWORK

Let's ask an LLM
for a generic definition,
just to get an idea
of the domain.

Trade Balance refers to the difference between a country's exports and imports over a given period. It's a key component of a nation's balance of payments.

When a country exports more than it imports, it has a trade surplus. When it imports more than it exports, it has a trade deficit.

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Okay, maybe II’ll also
ask to give me some examples
of possible scenarios,
so that I can understand it better.

Here are five example use-case scenarios of trade balance:

1. **Policy Development**: A country experiencing a persistent trade deficit might develop targeted policies to boost exports in specific sectors where they have comparative advantages, such as implementing tax incentives for exporters or investing in infrastructure that facilitates trade.

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S Give me a definition of "Trade Balance".

Okay, maybe II’ll also
ask to give me some examples
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so that I can understand it better.

S ok, now give me 5 example use-case scenarios of trade balance

That’s great! Could you
give me also some user story?
And maybe the
competency questions...

S Ok, now consider the scenario "Policy Development",
and give me a small user story for it

User Role

Goal

Outcome

User Story: Policy Development and Trade Balance

As a senior economic advisor to the Minister of Trade,

I want to analyze our country's persistent trade deficit in the technology sector,

So that I can develop targeted policy recommendations to improve our competitive position.

Context:

Our country has been experiencing a growing trade deficit in high-tech products over the past five years, despite having a well-educated workforce and established tech innovation hubs.

Actions:

- Review quarterly trade balance reports focusing on technology imports/exports
- Identify specific sub-sectors with the largest deficits
- Compare our tech sector's performance against regional competitors
- Analyze successful export promotion strategies from similar economies
- Consult with industry leaders about barriers to international competitiveness

Desired Outcome:

Develop a comprehensive policy proposal that includes tax incentives for tech exporters, targeted R&D funding, skills development programs, and potential regulatory changes that would help local tech companies increase their export capacity and reduce the sector's contribution to our overall trade deficit.

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Ok, now consider the scenario "Policy Development", and give me a small user story for it

Ok. Now
model it for me.

ok, now if you should make an ontology in RDF of the "trade balance" domain, what classes and properties would you create?

I'll create an RDF ontology for the "trade balance" domain with key classes and properties. Here's what this ontology would include:

Classes

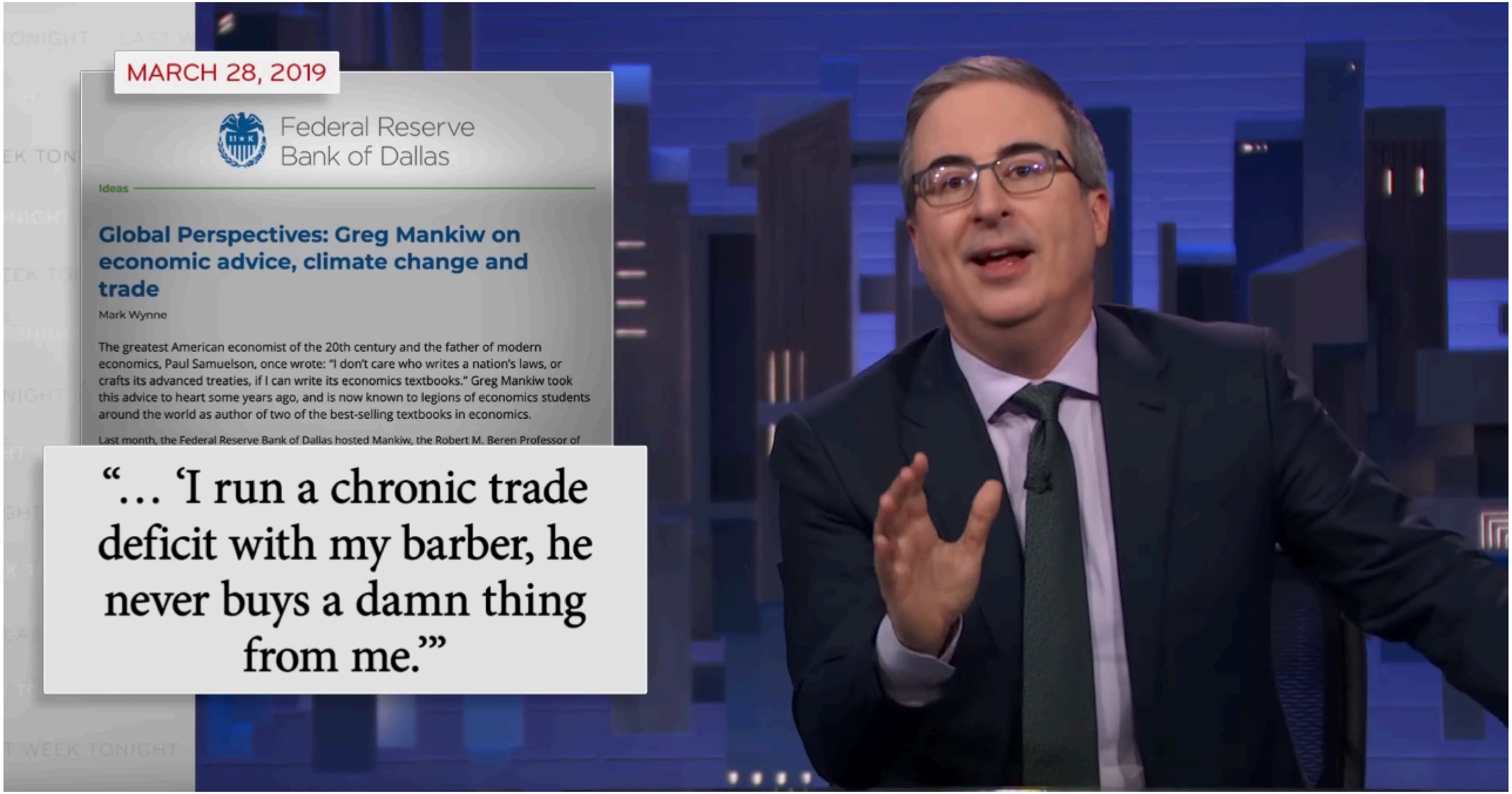
- `TradeBalance`: The central concept representing the difference between exports and imports
- `Country`: Nations engaging in international trade
- `TradeMetric`: Different measurements related to trade (e.g., exports, imports)
- `TradeGood`: Products and services being traded
- `EconomicSector`: Groupings of economic activities (manufacturing, services, etc.)
- `TimePeriod`: Temporal units for measuring trade
- `Currency`: Monetary units used in valuation
- `TradePolicy`: Government actions affecting trade
- `TradeAgreement`: Formal arrangements between nations
- `EconomicIndicator`: Broader economic measurements that influence trade

Object Properties

- `hasTradeBalance`: Links a country to its trade balance
- `hasTradePartner`: Connects countries in trading relationships
- `affectedBy`: Links trade balance to influencing factors
- `measuredIn`: Connects trade metrics to their currency
- `measuredFor`: Links trade metrics to time periods
- `belongsToSector`: Associates trade goods with economic sectors
- `implementsPolicy`: Connects countries to their trade policies
- `participatesIn`: Links countries to trade agreements
- `influences`: Shows causal relationships between indicators and trade

Data Properties

- `exportValue`: Numeric value of exports
- `importValue`: Numeric value of imports
- `balanceValue`: Numeric value of trade balance
- `tradeVolume`: Total value of combined exports and imports
- `timeFrame`: When the trade activity occurred
- `growthRate`: Percentage change over time
- `percentageOfGDP`: Trade metrics as percentage of national GDP
- `deficitThreshold`: Policy-defined acceptable deficit level
- `tariffRate`: Percentage tax on specific imports


MARCH 28, 2019
Federal Reserve Bank of Dallas
Ideas
Global Perspectives: Greg Mankiw on economic advice, climate change and trade
Mark Wynne
The greatest American economist of the 20th century and the father of modern economics, Paul Samuelson, once wrote: "I don't care who writes a nation's laws, or crafts its advanced treaties, if I can write its economics textbooks." Greg Mankiw took this advice to heart some years ago, and is now known to legions of economics students around the world as author of two of the best-selling textbooks in economics.
Last month, the Federal Reserve Bank of Dallas hosted Mankiw, the Robert M. Beren Professor of
"… 'I run a chronic trade deficit with my barber, he never buys a damn thing from me.'"

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Thanks for your attention.

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